

Host a Simple PHP Website Using Apache Virtual Hosting

Question

Host a simple PHP website in a user home directory.

1. Create a new user named Alice
2. Create a sample web page under a directory named public_html (/home/Alice/public_html)
3. Access the website using URL: myfirstwebsite.test and www.myfirstwebsite.test

Webpage content:

```
phpinfo();
```

```
?>
```

1. Objective

To host a simple PHP website from a user's home directory using **Apache name-based virtual hosting** and access it using `myfirstwebsite.test` and www.myfirstwebsite.test.

2. Environment

- Operating System: Red Hat Enterprise Linux 10.2
- Web Server: Apache HTTP Server
- PHP
- User: sanjid
- Website Directory: /home/sanjid/public_html

3. Procedure

Step 1: Install Apache and PHP

Apache HTTP Server and PHP were installed:

```
sudo dnf install httpd php -y
```

Apache was enabled and started:

```
sudo systemctl enable --now httpd
```

Step 2: Create the Website Directory

The `public_html` directory was created inside the user's home directory:

```
sudo mkdir -p /home/sanjid/public_html
```

Step 3: Create the PHP Webpage

The PHP webpage was created at:

```
/home/sanjid/public_html/index.php
```

The webpage contains:

```
<?php  
phpinfo();  
?>
```

The `phpinfo()` function displays detailed information about the PHP environment.

Step 4: Configure Permissions

The website directory was assigned to the `sanjid` user:

```
sudo chown -R sanjid:sanjid /home/sanjid/public_html
```

The required permissions were configured:

```
sudo chmod 755 /home/sanjid  
sudo chmod 755 /home/sanjid/public_html
```

Step 5: Configure Apache Virtual Hosting

A virtual host configuration file was created:

```
sudo vi /etc/httpd/conf.d/myfirstwebsite.conf
```

The following configuration was added:

```
<VirtualHost *:80>
    ServerName myfirstwebsite.test
    ServerAlias www.myfirstwebsite.test

    DocumentRoot /home/sanjid/public_html

    <Directory /home/sanjid/public_html>
        AllowOverride All
        Require all granted
    </Directory>

    DirectoryIndex index.php
</VirtualHost>
```

This configuration makes Apache serve the website from /home/sanjid/public_html for both myfirstwebsite.test and www.myfirstwebsite.test.

Step 6: Configure SELinux

Since SELinux was enforcing, the website directory was assigned the appropriate Apache web-content context:

```
sudo chcon -R -t httpd_sys_content_t /home/sanjid/public_html
```

Step 7: Restart Apache

Apache was restarted to apply the virtual host configuration:

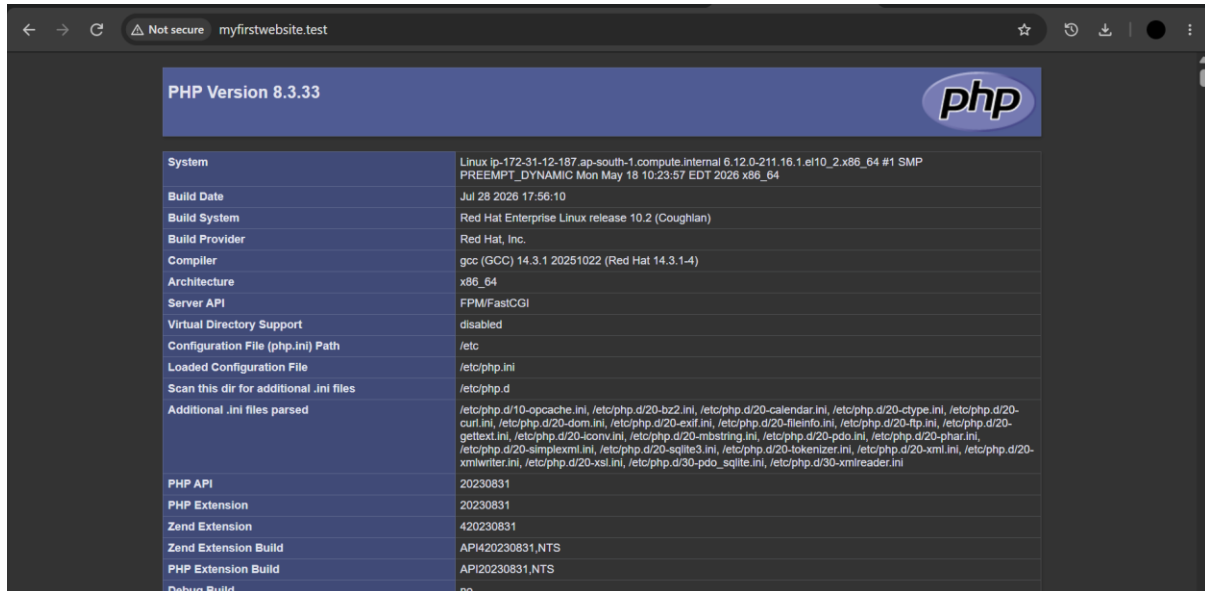
```
sudo systemctl restart httpd
```

Step 8: Configure Hostname Resolution

The domain names were mapped to the EC2 instance's public IP address in the client's hosts file:

15.207.14.213 myfirstwebsite.test www.myfirstwebsite.test

4. Output



PHP Version 8.3.33	
System	Linux ip-172-31-12-187-ap-south-1-compute.internal 6.12.0-211.16.1.el10_2.x86_64 #1 SMP PREEMPT_DYNAMIC Mon May 18 10:23:57 EDT 2026 x86_64
Build Date	Jul 28 2026 17:56:10
Build System	Red Hat Enterprise Linux release 10.2 (Coughlan)
Build Provider	Red Hat, Inc.
Compiler	gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-imagick.ini, /etc/php.d/20-ldap.ini, /etc/php.d/20-ldap3.ini, /etc/php.d/20-libsmbclient.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mcrypt.ini, /etc/php.d/20-mysqlnd.ini, /etc/php.d/20-odbc.ini, /etc/php.d/20-oci8.ini, /etc/php.d/20-openssl.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-pdo_dblib.ini, /etc/php.d/20-pdo_firebird.ini, /etc/php.d/20-pdo_oci.ini, /etc/php.d/20-pdo_odbc.ini, /etc/php.d/20-pdo_pgsql.ini, /etc/php.d/20-pdo_sqlite.ini, /etc/php.d/20-pgsql.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-redis.ini, /etc/php.d/20-session.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlrpc.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no

5. Result

The PHP website was successfully hosted using **Apache name-based virtual hosting** from:

`/home/sanjid/public_html`

The website successfully displayed the **PHP Info page** through the configured virtual host.

The configured URLs are:

<http://myfirstwebsite.test><http://www.myfirstwebsite.test>

6. Conclusion

The PHP website was successfully hosted from a user's home directory using Apache name-based virtual hosting. The virtual host was configured with a `ServerName` and `ServerAlias`, the required permissions and SELinux context were applied, and the PHP Info page was successfully displayed.